

THEORY AND METHODS

NIG-GARCH MODELS BASED ON OPEN, CLOSE, HIGH AND LOW PRICES

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Summary: Lildholdt (2002) introduces a method to fit GARCH models to financial return series using open, close, high and low prices. This method assumes that the price process on each day follows a lognormal diffusion with constant volatility for that day. It is well known that the consequent normality of the innovation distribution of the GARCH model for the return process is too light tailed to be realistic in many cases in practice. Accumulating evidence points to the NIG based approach being more realistic. This paper extends Lildholdt's model to incorporate the latter approach. An example illustrates the effectiveness of the extension.

1. Introduction

Consider the returns corresponding to the daily closing prices of a stock. It is a well-known and frequently observed phenomenon that such a return series is subject to volatility clustering. GARCH models provide an effective approach towards dealing with this phenomenon. The relative success of the GARCH approach is amply demonstrated by the huge growth in the literature on these models (see surveys by Bollerslev, Chou and Kroner (1992), Bollerslev,

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Engle and Nelson (1994), Ghysels, Harvey and Renault (1996) and Li, Ling and McAleer (2002)). GARCH models can be fitted to the return series by maximum likelihood. The returns are usually computed as the log of the closing price ratio on successive days. Quite often the daily open, high and low prices of the stock are available in addition to the closing prices. In principle better modelling should be possible using this additional data. The issue discussed in this paper is how to do this. Pre-GARCH contributions on this topic were made by Parkinson (1980), Garman and Klass (1980) and Ball and Torous (1984). This literature is reviewed by Lildholdt (2002) who then introduces a GARCH model based on the assumption that over each day anew the incremental log-price process follows a Brownian motion with constant volatility for that day. This enables him to write down the likelihood function for returns data based on open, close, high and low prices so that model fitting then becomes a maximum likelihood inference issue. A consequence of the Brownian motion assumption is that the innovation distribution of the GARCH model is normal. It is now well known that the normal distribution is often too light-tailed to be realistic in this context, especially when dealing with high frequency (such as daily) data. An early alternative is the t-distribution introduced by Bollerslev (1987); more recently the normal inverse Gaussian (NIG) has become a strong favourite (see e.g. Barndorff-Nielsen and Prause (2001), Jensen and Lunde (2001), Forsberg (2002), Forsberg and Bollerslev (2002), Venter and de Jongh (2003), etc.). The aim of this paper is to extend the work of Lildholdt (2002) to incorporate the NIG alternative.

The paper is organised as follows. In Section 2 we review briefly some aspects of inverse Gaussian (IG), generalized inverse Gaussian (GIG) and NIG distributions required here; in particular, we introduce the notions of a unit IG

distribution and a standard NIG distribution. In Section 3 these notions enable us to formulate NIG-GARCH models concisely for the univariate case when only the returns based on closing prices are used. In Section 4 we extend the work of Lildholdt (2002) to the NIG-GARCH model when returns based on open, high, low and closing prices are used. In Section 5 we illustrate with an example and in Section 6 we close with pertinent remarks.

2. GIG, IG and NIG distributions and their properties

The generalized inverse Gaussian distribution with parameters $(\lambda, \delta, \gamma)$ has density

$$f_{GIG(\lambda, \delta, \gamma)}(w) = \left(\frac{\gamma}{\delta}\right)^\lambda \frac{w^{\lambda-1}}{2K_\lambda(\delta\gamma)} \exp\left[-\frac{1}{2}(\delta^2 w^{-1} + \gamma^2 w)\right] \quad (2.1)$$

for $w > 0$, where $\delta, \gamma > 0$

and K_γ is the modified Bessel function of third order (Jorgensen, 1982). We refer to this as the $GIG(\lambda, \delta, \gamma)$ distribution. If W is $GIG(\lambda, \delta, \gamma)$ distributed,

$$EW^k = \delta^k K_{\lambda+k}(\delta\gamma) / \gamma^k K_\lambda(\delta\gamma) \quad (2.2)$$

since (2.1) integrates to 1. For the special case $\lambda = -\frac{1}{2}$ we get the inverse Gaussian distribution (referred to as the $IG(\delta, \gamma)$ distribution). Taking $\delta = \gamma = \psi > 0$, leads to what we shall call the unit inverse Gaussian distribution (denoted by $UIG(\psi)$) with density

$$\begin{aligned} f_{UIG(\psi)}(w) &= \frac{1}{2K_{\frac{1}{2}}(\psi^2)} w^{-\frac{3}{2}} \exp\left[-\frac{1}{2}\psi^2(w^{-1} + w)\right] \\ &= \frac{\psi \exp(\psi^2)}{\sqrt{2\pi}} w^{-\frac{3}{2}} \exp\left[-\frac{1}{2}\psi^2(w^{-1} + w)\right] \quad \text{for } w > 0. \end{aligned} \quad (2.3)$$

If W is $UIG(\psi)$ -distributed then $EW^k = K_{k-\frac{1}{2}}(\psi^2)/K_{\frac{1}{2}}(\psi^2)$ and in particular

$$EW = 1 \text{ while } Var(W) = 1/\psi^2. \quad (2.4)$$

The fact that $EW = 1$ regardless of the value of $\psi > 0$ provides the reason for calling this the “unit” IG distribution. It is easy to see that if W is $UIG(\psi)$ distributed then $V = \delta W/\gamma$ is $IG(\delta, \gamma)$ distributed for any $\delta, \gamma > 0$ subject to $\sqrt{\delta\gamma} = \psi$.

Suppose Z is $N(0, 1)$ distributed and W is $UIG(\psi)$ distributed independently of Z . Then we call the distribution of $X = \beta W + \sqrt{W}Z$ the standard normal inverse Gaussian distribution with parameters β (any real number) and $\psi > 0$, abbreviated as $SNIG(\beta, \psi)$. To calculate the density function, note that the conditional density of X given $W = w$ is $\varphi((x - \beta w)/\sqrt{w})/\sqrt{w}$ with φ the $N(0, 1)$ -density. Consequently

$$\begin{aligned} f_{SNIG(\beta, \psi)}(x) &= \int_0^\infty \frac{1}{\sqrt{w}} \varphi\left(\frac{x - \beta w}{\sqrt{w}}\right) f_{UIG(\psi)}(w) dw \\ &= \frac{\psi}{\pi} \left(\frac{\beta^2 + \psi^2}{x^2 + \psi^2}\right)^{\frac{1}{2}} e^{\psi^2 + \beta x} K_1\left(\sqrt{\beta^2 + \psi^2} \sqrt{x^2 + \psi^2}\right). \end{aligned} \quad (2.5)$$

In terms of the usual $(\alpha, \beta, \mu, \delta)$ parameter specification of the NIG family (see e.g. Barndorff-Nielsen et al., 1985 or Venter and de Jongh, 2003), (2.5) corresponds to the choice $(\sqrt{\beta^2 + \psi^2}, \beta, 0, \psi)$. The full four-parameter NIG family can be obtained by translation and scale transforms of the SNIG distributions. Moments of the SNIG distribution follow easily from the defining relation $X = \beta W + \sqrt{W}Z$.

3. NIG-GARCH models

Let $Y_1, Y_2, \dots$ denote the daily returns of a stock and suppose we propose to describe it in terms of a GARCH model of the form

$$Y_n = \mu_n + \sqrt{h_n} X_n \quad \text{for } n = 1, 2, \dots \quad (3.1)$$

where μ_n represents an expected (or structural) component, $\sqrt{h_n}$ is the volatility and X_n the innovation at time n . It is assumed that μ_n and h_n are at most dependent on the past information set $\mathcal{F}_{n-1}$ (i.e. the σ -field generated by $Y_1, Y_2, \dots, Y_{n-1}$). Typical choices are $\mu_n = v + \phi Y_{n-1}$ and $h_n = \alpha_0 + \alpha_1 (Y_{n-1} - \mu_{n-1})^2 + \beta_1 h_{n-1}$ which yield the popular AR(1)-GARCH(1,1) model. The innovations X_n are supposed to be independent of $\mathcal{F}_{n-1}$ and also independent and identically distributed (iid) with common (innovation) density g . To make μ_n and h_n identifiable, we assume that $EX_n = 0$ and $Var(X_n) = 1$ so that $\mu_n = E[Y_n | \mathcal{F}_{n-1}]$ and $h_n = Var[Y_n | \mathcal{F}_{n-1}]$. Often, g is assumed to be the $N(0, 1)$ density in which case (3.1) may be written as

$$Y_n = \mu_n + \sqrt{h_n} Z_n \quad (3.2)$$

where $Z_1, Z_2, \dots$ are iid $N(0, 1)$. If this model is fitted and corresponding residuals are calculated and tested against the normality assumption by a normal QQ plot or a normal probability integral transform (PIT) plot, it often appears that an innovation distribution with heavier tails are required (see e.g. Forsberg, 2002). Recently a number of authors have argued in favour of using the NIG distribution for this purpose; see e.g. Andersson (2001), Jensen and Lunde (2001), Venter and de Jongh (2002, 2003), Forsberg (2002) and especially the important paper of Forsberg and Bollerslev (2002). This has led to what may be termed the NIG-GARCH model, which may be formulated as follows using the terminology of Section 2. Let $W_1, W_2, \dots$ be iid $UIG(\psi)$

distributed independently of $Z_1, Z_2, \dots$ and take $X_n = \sqrt{W_n}Z_n$ so that (3.1) may be written as

$$Y_n = \mu_n + \sqrt{h_n} \left(\sqrt{W_n}Z_n \right) = \mu_n + \sqrt{h_n W_n}Z_n. \quad (3.3)$$

From Section 2 we know that $\sqrt{W_n}Z_n$ has a $SNIG(0, \psi)$ distribution, so that the first expression in (3.3) differs from the model in (3.2) in the replacement of the $N(0, 1)$ -distributed innovation by a $SNIG(0, \psi)$ innovation. The second expression in (3.3) indicates that we may interpret this model alternatively as keeping the $N(0, 1)$ innovation but adjusting the squared volatility h_n inherited from the past by the random factor W_n . Since $EW_n = 1$ there is no adjustment on average but on any particular day the adjustment may be up or down as reflected by the value of W_n . Clearly, in cases where today's volatility may be affected by factors (arrival of news) operating after close of the market yesterday, it would be more realistic to introduce such an adjustment factor to cater for the effects which could not have been incorporated in yesterday's volatility measurement. This is precisely what the model (3.3) accomplishes.

More generally, we could take $X_n = \beta W_n + \sqrt{W_n}Z_n$ so that (3.3) is generalized to

$$Y_n = \mu_n + \sqrt{h_n} \left(\beta W_n + \sqrt{W_n}Z_n \right) = \mu_n + \beta \sqrt{h_n}W_n + \sqrt{h_n W_n}Z_n \quad (3.4)$$

thus also allowing for a possible shift impact of the adjustment factor on the structural part of the model. This model is comparable to a GARCH-in-mean model which would have the term $\beta \sqrt{h_n}W_n$ replaced by $\beta \sqrt{h_n}$. Again, since $EW_n = 1$, on average these terms are equal but one can argue that if volatility impacts on the structural part of the model then allowance should be made to incorporate into the structural part the random factor W_n due to effects of factors operating after close of the market on the previous day, so

that (3.4) should yield a more realistic formulation which may be described as a random GARCH-in-mean model.

Writing (3.4) in the form $Y_n = \mu_n + \sqrt{h_n}X_n$ with $X_n = \beta W_n + \sqrt{W_n}Z_n$ which is $SNIG(\beta, \psi)$ distributed according to the previous section, the log-likelihood function may be written as

$$\sum_n \left\{ \log \left[f_{SNIG(\beta, \psi)} \left((Y_n - \mu_n) / \sqrt{h_n} \right) \right] - \frac{1}{2} \log(h_n) \right\} \quad (3.5)$$

so that standard maximum likelihood (ML) methodology is available to estimate the parameters of the model. Once we have estimates $\hat{\beta}$, $\hat{\psi}$ and $\hat{\mu}_n$, $\hat{h}_n$ we can compute daily return residuals $\hat{X}_n = (Y_n - \hat{\mu}_n) / \sqrt{\hat{h}_n}$ and compare them against a $SNIG(\hat{\beta}, \hat{\psi})$ distribution using QQ or PIT plots to check the fit of the model.

The volatility adjustment factors W_n are unobserved (latent) variables which may be estimated from the observed returns $Y_1, Y_2, \dots, Y_n$ by its minimum mean squared error estimator $E[W_n | \mathcal{F}_n]$ as follows. From (3.4) the joint density of Y_n and W_n given $\mathcal{F}_{n-1}$ is $\varphi(y - \mu_n - \sqrt{h_n}\beta w) / \sqrt{h_n w} f_{UIG}(w) / \sqrt{h_n w}$. Consequently

$$\begin{aligned} E[W_n | \mathcal{F}_n] &= \frac{\int_0^\infty w \frac{1}{\sqrt{h_n w}} \varphi\left(\frac{Y_n - \mu_n - \sqrt{h_n}\beta w}{\sqrt{h_n w}}\right) f_{UIG}(w) dw}{\int_0^\infty \frac{1}{\sqrt{h_n w}} \varphi\left(\frac{Y_n - \mu_n - \sqrt{h_n}\beta w}{\sqrt{h_n w}}\right) f_{UIG}(w) dw} \\ &= \frac{\delta_n K_0(\gamma \delta_n)}{\gamma K_1(\gamma \delta_n)} \end{aligned} \quad (3.6)$$

where $\delta_n = \sqrt{(Y_n - \mu_n)^2 / h_n + \psi^2}$ and $\gamma = \sqrt{\beta^2 + \psi^2}$. This expression still depends on the parameters for which we may substitute their ML estimates yielding the estimate $\hat{W}_n$ of W_n . Although this estimate is the best we can do from a mean squared error point of view, we cannot expect it to be very

accurate generally since it largely depends on the information in the single observation Y_n . Indeed a simulation study revealed that its R^2 value may be as low as 10% in representative cases. Additional data on the price movements on each day should be useful to improve on this situation and the next section studies the case where such data come in the form of high and low prices in addition to open and close prices.

4. Extension of the NIG-GARCH model to incorporate open, high, low and close prices

To extend the NIG-GARCH model of the previous section to incorporate open, high, low and close prices, requires additional notation. Let $p_{t,n}$ denote the price of the stock at time t on day n where we measure time in terms of a fraction of a trading day (so $0 \leq t \leq 1$, with $t = 0$ corresponding to opening time and $t = 1$ to closing time). Also let $Y_{t,n} = \log(p_{t,n}) - \log(p_{0,n})$ denote the return on day n up to time t . Let W_n be as above and $Z_{t,n}$ be standard Brownian motions in t , independent over n and also independent of the W_n 's. Also introduce the innovation process $X_{t,n} = \beta t W_n + \sqrt{W_n} Z_{t,n}$, $0 \leq t \leq 1$. Following and extending the work of Lildholdt (2002), we then model $Y_{t,n}$ by

$$Y_{t,n} = t\mu_n + \sqrt{h_n} X_{t,n} = \left(\mu_n + \beta \sqrt{h_n} W_n \right) t + \sqrt{h_n W_n} Z_{t,n}.$$

If W_n were constant with value 1 then conditionally given the information before day n , the return process within day n would follow a Wiener process with drift rate and volatility inherited from the past via μ_n and h_n respectively. However, with W_n stochastic, the inherited drift rate and volatility are allowed to be adjusted by random factors affecting the return process after close on the previous day. Notice, however, that W_n does not change with time on day n so that within each day this model does not allow

further stochastic volatility effects. The model can be generalized further in this direction but we shall not do so in this paper.

Let p_n^o , p_n^h , p_n^l , p_n^c denote the open, high, low and close prices of the stock on day n . Then the **total return** on day n is

$$\begin{aligned} Y_n &= \log(p_n^c) - \log(p_n^o) = Y_{1,n} \\ &= \mu_n + \beta\sqrt{h_n}W_n + \sqrt{h_nW_n}Z_{1,n} \end{aligned} \quad (4.2)$$

which clearly amounts to the model of Section 3 for daily returns. The high price on day n may be transformed to the **high return**

$$\begin{aligned} A_n &= \log(p_n^h) - \log(p_n^o) = \max \{ \log(p_{t,n}) - \log(p_{0,n}) : 0 \leq t \leq 1 \} \\ &= \max \{ Y_{t,n} : 0 \leq t \leq 1 \} \\ &= \max \left\{ \left(\mu_n + \beta\sqrt{h_n}W_n \right) t + \sqrt{h_nW_n}Z_{t,n} : 0 \leq t \leq 1 \right\} \end{aligned} \quad (4.3)$$

and the low price on day n may be transformed to the **low return**

$$\begin{aligned} B_n &= \log(p_n^o) - \log(p_n^l) \\ &= -\min \{ \log(p_{t,n}) - \log(p_{0,n}) : 0 \leq t \leq 1 \} \\ &= -\min \{ Y_{t,n} : 0 \leq t \leq 1 \} \\ &= -\min \left\{ \left(\mu_n + \beta\sqrt{h_n}W_n \right) t + \sqrt{h_nW_n}Z_{t,n} : 0 \leq t \leq 1 \right\}. \end{aligned}$$

To be able to calculate the likelihood function from the total, high and low return data we need the joint density function of (Y_n, A_n, B_n) conditionally given the past information set $\mathcal{F}_{n-1}$ (now the σ -field generated by (Y_m, A_m, B_m) , $m \leq n-1$). To calculate this density, let U_t denote a Wiener process with drift rate μ and volatility σ , let $A = \max \{U_t : 0 \leq t \leq 1\}$, $B = -\min \{U_t : 0 \leq t \leq 1\}$ and let $f(y, a, b | \mu, \sigma^2)$ denote the joint density of (U_1, A, B) in the argument (y, a, b) . Then the conditional density of (Y_n, A_n, B_n) given $\mathcal{F}_{n-1}$ and also $W_n = w$ is given by $f(y, a, b | \mu_n + \beta\sqrt{h_n}w, h_nw)$ so that the

conditional density of (Y_n, A_n, B_n) given only $\mathcal{F}_{n-1}$ may be expressed as

$$p(y, a, b | \mu_n, h_n) = \int_0^\infty f(y, a, b | \mu_n + \beta \sqrt{h_n} w, h_n w) f_{UIG(\varphi)}(w) dw. \quad (4.5)$$

In order to calculate this integral we need an expression for $f(y, a, b | \mu, \sigma^2)$. One possible expression is given on page 25 of Lildholdt (2002), obtained by differentiating equation (81) on page 222 of Cox and Miller (1965) with respect to a and b . We have greater success in proceeding from the alternative equation (78) on page 222 of Cox and Miller (1965). Differentiating the latter with respect to a and b we obtain

$$\begin{aligned} f(y, a, b | \mu, \sigma^2) &= \frac{4}{\sqrt{2\pi}\sigma^5} \exp\left(-(y-\mu)^2/2\sigma^2\right) \times \\ &\quad \sum_{m=-\infty}^{\infty} \left[m^2 \left\{ (y-y'_m)^2 - \sigma^2 \right\} \exp\left\{-(y'^2_m - 2y'_m y)/2\sigma^2\right\} - \right. \\ &\quad \left. m(m-1) \left\{ (y-y''_m)^2 - \sigma^2 \right\} \exp\left\{-(y''^2_m - 2y''_m y)/2\sigma^2\right\} \right] \quad (4.6) \end{aligned}$$

where $y'_m = 2m(a+b)$ and $y''_m = 2a - y'_m$ for $m = 0, \pm 1, \pm 2, \dots$. Substituting this expression into (4.5), carrying out the integration and simplifying the result lead to the expression

$$\begin{aligned} p(y, a, b | \mu_n, h_n) &= \frac{2}{\pi h_n^{5/2}} \psi \exp\left(\psi^2 + \beta(y - \mu_n)/\sqrt{h_n}\right) \times \\ &\quad \sum_{m=-\infty}^{\infty} \left[m^2 \left\{ (y - y'_m)^2 (\beta_\psi/\psi'_{mn})^3 K_3(\beta_\psi \psi'_{mn}) \right. \right. \\ &\quad \left. \left. - h_n (\beta_\psi/\psi'_{mn})^2 K_2(\beta_\psi \psi'_{mn}) \right\} - \right. \\ &\quad \left. m(m-1) \left\{ (y - y''_m)^2 (\beta_\psi/\psi''_{mn})^3 K_3(\beta_\psi \psi''_{mn}) \right. \right. \\ &\quad \left. \left. - h_n (\beta_\psi/\psi''_{mn})^2 K_2(\beta_\psi \psi''_{mn}) \right\} \right] \quad (4.7) \end{aligned}$$

where

$$\begin{aligned}\beta_\psi^2 &= \beta^2 + \psi^2 \\ \psi'_{mn} &= \left[\psi^2 + \left\{ (y - \mu_n)^2 + y_m^2 - 2y'_m y \right\} / h_n \right]^{\frac{1}{2}} \\ \psi''_{mn} &= \left[\psi^2 + \left\{ (y - \mu_n)^2 + y_m'^2 - 2y''_m y \right\} / h_n \right]^{\frac{1}{2}}\end{aligned}\quad (4.8)$$

and where K_2, K_3 denote the modified Bessel functions of third order and indices 2 and 3 respectively. It is readily shown that $y_m'^2 - 2y'_m y, y_m''^2 - 2y''_m y > 0$ for all cases which are required to make the integration leading to (4.7) valid and imply the existence of the square roots in (4.8). The term $m = 0$ does not contribute to the sum in (4.7) while only the first part contributes when $m = 1$. We compute (4.7) by summing separately over the positive and negative values of m until convergence in each direction is achieved. Typically this requires fewer than 10 terms in each range. The log-likelihood function of the $\{Y_n, A_n, B_n\}$ data may now be written in terms of (4.7) as

$$\sum_n \log p(Y_n, A_n, B_n | \mu_n, h_n). \quad (4.9)$$

Again, standard maximum likelihood inference may now be used to fit this extended NIG-GARCH model. We developed a program based on Proc NLP of SAS to compute MLE estimates and corresponding asymptotic standard errors. This is illustrated in Section 5.

As to model checking, we can again compute total return residuals $\hat{X}_n = (Y_n - \hat{\mu}_n) / \hat{h}_n$ and then compare them against a $SNIG(\hat{\beta}, \hat{\psi})$ distribution as in Section 3. We may also wish to do model checking in terms of the high and low returns. To see how this might be approached, introduce the residual process $\hat{X}_{t,n} = (Y_{t,n} - t\hat{\mu}_n) / \sqrt{\hat{h}_n}, 0 \leq t \leq 1$ which approximates the process $X_{t,n} = \beta t W_n + \sqrt{W_n} Z_{t,n}, 0 \leq t \leq 1$. Define $\hat{A}_n = \max \{ \hat{X}_{t,n} : 0 \leq t \leq 1 \}$ which approximates $A_n^X =$

$\max \{X_{t,n} : 0 \leq t \leq 1\}$ and $\hat{B}_n = -\min \{\hat{X}_{t,n} : 0 \leq t \leq 1\}$ which approximates $B_n^X = -\min \{X_{t,n} : 0 \leq t \leq 1\}$. Then the joint distribution of $\{\hat{X}_{1,n}, \hat{A}_n, \hat{B}_n\}$ would be approximately that of $\{X_{1,n}, A_n^X, B_n^X\}$ which has the density $p(y, a, b | 0, 1)$ given by (4.7). From this we can derive PIT plots to evaluate the fit of the $\hat{A}_n$'s and $\hat{B}_n$'s separately and also that of the relevant joint distributions. Unfortunately we need the full intra-day returns process data $Y_{t,n}, 0 \leq t \leq 1$, to be able to calculate $\hat{A}_n$ and $\hat{B}_n$. In general there is no way to calculate $\hat{A}_n$ and $\hat{B}_n$ directly from A_n and B_n (or the available high, low and open prices). In the absence of full intra-day data, this residual analysis approach is generally not feasible. One special case where it is indeed feasible is when we may assume that $\mu_n = 0$. Then $\hat{\mu}_n = 0$, $\hat{X}_{t,n} = Y_{t,n}/\sqrt{\hat{h}_n}$, $\hat{A}_n = A_n/\sqrt{\hat{h}_n}$, $\hat{B}_n = B_n/\sqrt{\hat{h}_n}$ and we do not need any further data to be able to proceed as outlined above.

Also, we can compute estimates $\widehat{W}_n$ of W_n using the additional information provided by the high and low returns as follows. Arguing as above, the joint density of (Y_n, A_n, B_n, W_n) given $\mathcal{F}_{n-1}$ is given by $f(y, a, b | \mu_n + \beta\sqrt{h_n}w, h_nw) f_{UIG}(w)$ so that

$$\begin{aligned} E[W_n | \mathcal{F}_n] &= \frac{\int_0^\infty w f(Y_n, A_n, B_n | \mu_n + \beta\sqrt{h_n}w, h_nw) f_{UIG}(\psi)(w) dw}{\int_0^\infty f(Y_n, A_n, B_n | \mu_n + \beta\sqrt{h_n}w, h_nw) f_{UIG}(w) dw} \\ &= \frac{q(Y_n, A_n, B_n | \mu_n, h_n)}{p(Y_n, A_n, B_n | \mu_n, h_n)}. \end{aligned} \quad (4.10)$$

The denominator p is given by (4.7) and the numerator q may be similarly expressed as

$$\begin{aligned}
q(y, a, b | \mu_n, h_n) = & \frac{2}{\pi h_n^{5/2}} \psi \exp(\psi^2 + \beta(y - \mu_n) / \sqrt{h_n}) \times \\
& \sum_{m=-\infty}^{\infty} \left[m^2 \left\{ (y - y'_m)^2 (\beta_\psi / \psi'_{mn})^2 K_2(\beta_\psi \psi'_{mn}) \right. \right. \\
& \quad \left. \left. - h_n (\beta_\psi / \psi'_{mn}) K_1(\beta_\psi \psi'_{mn}) \right\} \right. \\
& \quad \left. - m(m-1) \left\{ (y - y''_m)^2 (\beta_\psi / \psi''_{mn})^2 K_2(\beta_\psi \psi''_{mn}) \right. \right. \\
& \quad \left. \left. - h_n (\beta_\psi / \psi''_{mn}) K_1(\beta_\psi \psi''_{mn}) \right\} \right]
\end{aligned} \tag{4.11}$$

Replacing the parameters in these expressions by their ML estimators we obtain the required $\widehat{W}_n$'s. Due to the additional information used here, these estimates are substantially more accurate than those of the previous section. Simulation studies show that R^2 values of the order of 50% can be obtained in representative cases.

Finally we note that the W_n 's are all equal to 1 in the special case $\psi = \infty$; if we also take $\beta = 0$ then (4.1) simply becomes $Y_{t,n} = t\mu_n + \sqrt{h_n}Z_{t,n}$ which is essentially the model treated by Lildholdt (2002).

5. Example

In this section we illustrate the results in terms of the IBM share price series over the period 1999/02/09 – 2003/01/31. We purchased IBM intra-day data from Tick Data, Inc and use the open, high, low and closing prices of the last 1000 days of our record. The closing prices are shown in Figure 1. The open, high and low prices follow these so closely that they cannot be distinguished from the closing prices on the graph. There is no observable long term trend in the price series over this period.

We fitted three AR(1)-GARCH(1,1) models to the data, namely

1. using the total return data Y_n only with the NIG model (3.4),
2. using the total, high and low return data $\{Y_n, A_n, B_n\}$ with the NIG

model of Section 4,

3. using the $\{Y_n, A_n, B_n\}$ data with the model of Lildholdt (2002).

Table 1 gives the ML estimates of the parameters involved in these models together with their asymptotic standard errors. If we take the standard errors into account we see that the estimates of the two NIG models agree reasonably well. However, as is to be expected, the standard errors of the second model are all smaller than those of the first model. This reflects the efficiency gain obtained from using the high and low series in addition to the closing prices. We note for example that β , ν and ϕ would not be considered significantly different from 0 according to the estimates based only on the closing prices but both β and ν are indeed significantly different from 0 according to the estimates based on the additional data. Again, the persistence parameter $\alpha_1 + \beta_1$ would not be considered significantly different from 1 according to the estimate based on the first model, which implies that we would have to consider the (somewhat uncomfortable) conclusion that an IGARCH model may be required. However, the estimate from the second model effectively eliminates this concern. The estimates of the AR and GARCH parameters from the third model also agree quite well with those of the first two models. This is in line with the well-known result that ML estimates based on normality are asymptotically consistent (Bollerslev and Wooldridge, 1992 and Gouriéroux, 1997). However, according to the second model the standard errors of the estimates of the two NIG parameters ψ and β suggest that these are indeed required and that we may expect this model to fit better than the normal model.

Figure 2 shows the PIT-plots of the daily return residuals corresponding to the three models. It is evident from the plots of the first two models that the NIG based approach leads to models whose residuals agree better with the model assumptions than for the third model. Further testing (Durbin-Watson, Q- and

LM-tests) on the residuals of the second model also indicates that they conform to the requirements of independence with no remaining GARCH effects.

We also computed the estimated volatility adjustment factors $\widehat{W}_n$ according to the first two models. Figures 3 and 4 show the results over a period of the series when volatility varied substantially. In Figure 3 the $\widehat{W}_n$'s are those of the first model which uses only the total daily returns and in Figure 4 the $\widehat{W}_n$'s are from the second model which uses the high and low returns as well. For comparison purposes we also show the daily realized volatilities in both figures. The notion of realized volatility has been studied extensively in recent years, both empirically and analytically (see e.g. Andersen et al., 2000-2002, Barndorff-Nielsen, O.E. and Shephard, N., 2002a,b). Basically it is a well justified measure of volatility that is computed as the sum of squares of returns over short intra-day periods. Figure 3 shows that the $\widehat{W}_n$'s based on the total daily returns adjust the Garch volatilities $\widehat{h}_n$ slightly towards the realized volatilities while Figure 4 shows the adjustment is much larger when the $\widehat{W}_n$'s incorporate the high and low returns as well. Keep in mind that in our example we had full intra-day data and were able to compute high and low returns as well as the realized volatilities, enabling us to carry out the above comparison. The results show that in cases where we have the open, close, high and low prices (but not full intra-day prices) we can still compute a volatility measure $\widehat{h}_n \widehat{W}_n$ that goes some way towards approximating the realized volatilities.

For daily returns the NIG based model has a $SNIG(\widehat{\beta}, \widehat{\psi})$ estimated innovation distribution while the normal model has $N(0, 1)$ -distributed innovations. These two innovation densities are shown in Figure 5. At first sight they may not appear much different, but their quantiles can differ substantially. For example, the 1% value at risk (VaR or quantile) is -2.88 for the NIG model and -2.33 for the normal model, a difference of about 24% for

the NIG model relative to the normal model. In risk applications the models may therefore lead to substantially different conclusions.

The estimates of ϕ suggest that the autoregressive part of the AR(1) component is not needed and one can refit the models with $\phi = 0$ to begin with. The resulting estimates agree quite closely with those of Table 2 and since no new matters arise we omit the details.

6. Concluding remarks

Two recent developments in the area of GARCH-type volatility modelling are as follows. On the one hand limited historical intra-day data in the form of open, high, low and close prices are often available and a method has been developed by Lildholdt (2002) to use all of these data rather than only close prices when fitting GARCH models. On the other hand the traditional lognormal diffusion approach used in GARCH modelling of return series based on close prices, has been shown to produce ill-fitting innovation distributions, but this problem can be overcome to a large extent by using NIG models instead. In this paper we have shown how these two developments may be combined to produce better fitting GARCH models for return data based on open, high, low and closing prices. In the process we introduced the notion of volatility adjustment latent variables, which clarify the role of the NIG innovation distribution and also help to adjust GARCH volatility estimates towards realized volatilities.

Finally, a comment on the decomposition used in our development. We assume that a typical innovation X can be decomposed in the form $X = \beta W + \sqrt{W}Z$ where Z is $N(0, 1)$ distributed and we modelled W as $UIG(\psi)$ distributed. This means that X has a normal mean and variance mixture distribution. A large advantage of using the $UIG(\psi)$ mixing

distribution here is that most of the resulting distributions and expressions can be calculated explicitly (in terms of Bessel functions). A relevant question is whether other mixing distributions can be used and in particular since the t -innovation distribution is quite common in GARCH modelling, whether a mixing distribution can be used that will produce a t -innovation distribution. The answer is affirmative: starting with a “unit inverse gamma” distribution for W we may develop a theory parallel to that set out above. If we take the special choice $\beta = 0$ the resulting innovation density becomes the scaled t -density but this does not fit as well as the general case where we allow $\beta \neq 0$. The distributions resulting in the latter case are again expressible in terms of somewhat more involved Bessel functions but when applied to the example of Section 4 the results largely agree. More details are given in Venter, de Jongh and Griebenow (2004).

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Table 1. ML Estimates for IBM Data

Parameter	Model (1): NIGy	Model (2): NIGyab	Model (3): NORyab
ψ (NIG steepness)	2.1547 (0.7127)	1.7420 (0.0937)	—
β (NIG skewness)	-0.1626 (0.1883)	-0.2236 (0.0688)	—
ν (ar intercept)	0.0025 (0.0037)	0.0035 (0.0011)	.0024 (0.0006)
δ (ar coefficient)	-0.0252 (0.0331)	-0.0267 (0.0302)	-0.0104 (0.0350)
α_0 (arch0) $\times 10^5$	0.8008 (0.5370)	1.3437 (0.4013)	1.4419 (0.4020)
α_1 (arch1)	0.0497 (0.0185)	0.0804 (0.0118)	0.0718 (0.0116)
β_1 (garch1)	0.9308 (0.0261)	0.8795 (0.0192)	0.8736 (0.0219)
$\alpha_1 + \beta_1$ (persistence)	0.9805 (0.0134)	0.9599 (0.0105)	0.9454 (0.0128)

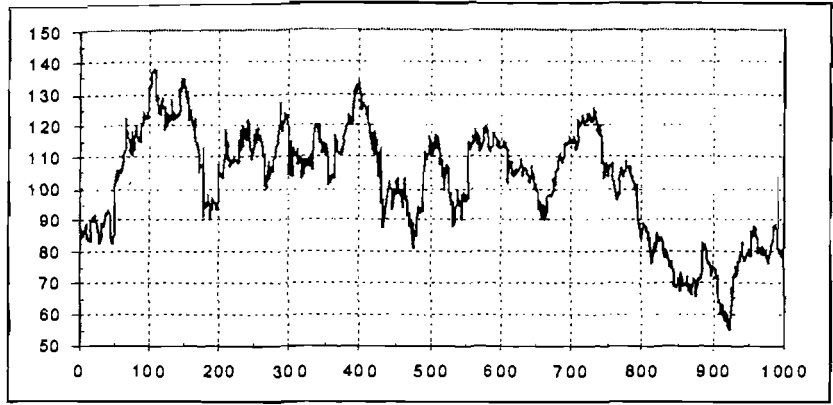


Figure 1. IBM Daily Closing Prices (1999/02/09 – 2003/01/31)

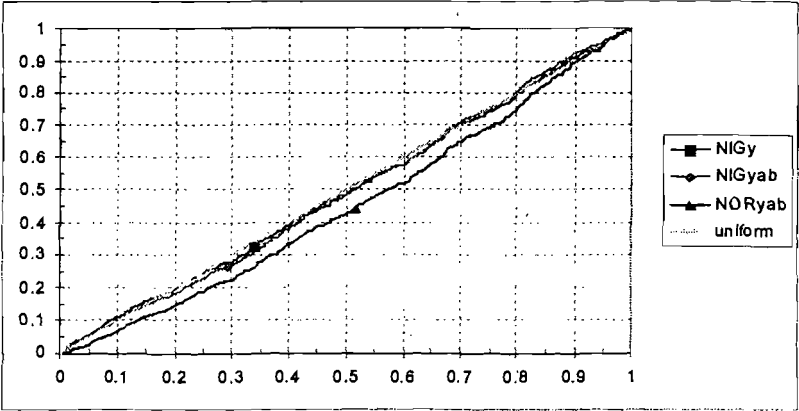


Figure 2. PIT Plots of Residuals

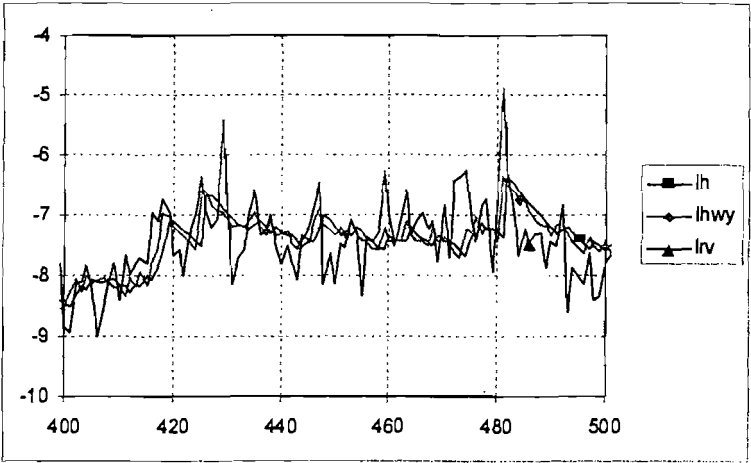


Figure 3. Comparing $\hat{h}_n$, realized volatility and $\hat{h}_n \hat{W}_n$ with $\hat{W}_n$ based on total returns (logarithm scales)

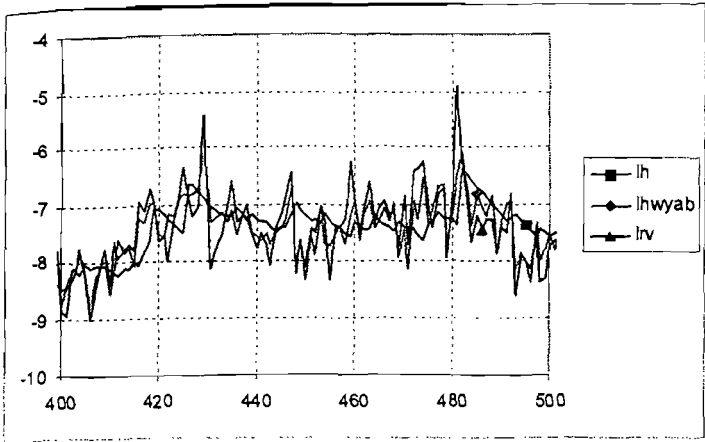


Figure 4. Comparing $\hat{h}_n$, realized volatility and $\hat{h}_n \hat{W}_n$ with $\hat{W}_n$ based on total, high and low returns (logarithmic scale)

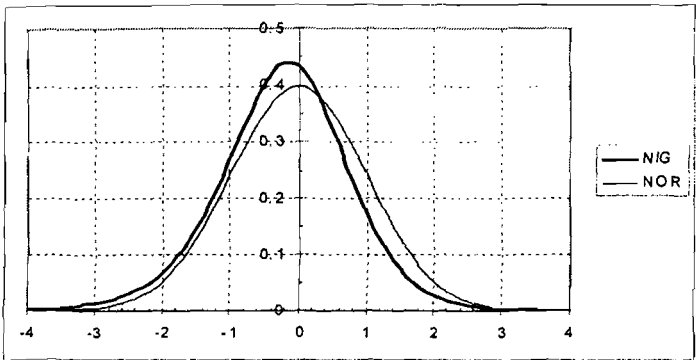


Figure 5. Innovation Distributions of NIG and Normal Models